

Design and Simulation of Planar and Gate-All-Around N-Channel JFETs for Low-Power Analog Applications

S. Sahil, T. Prajwal, and Manasi Nerurkar

Abstract—This letter presents a comparative TCAD Sentaurus simulation study of three n-channel junction field-effect transistor (JFET) architectures: a long-channel planar JFET, a geometrically scaled planar JFET, and a gate-all-around (GAA) JFET. Using consistent geometry definitions, doping profiles, meshing strategies, and physical models, the impact of device scaling and gate geometry on DC electrical characteristics is systematically analyzed. Transfer and output characteristics are evaluated with particular emphasis on transconductance, off-state leakage current, and threshold voltage behavior. While planar device scaling enhances transconductance, it also results in increased leakage due to degraded electrostatic control. In contrast, the GAA JFET exhibits superior gate control through uniform channel depletion, leading to higher transconductance, reduced leakage current, and improved threshold voltage stability. These results highlight the advantages of GAA JFET architectures for low-power and low-noise analog circuit applications.

Index Terms—JFET, gate-all-around (GAA), TCAD Sentaurus, device scaling, low-power analog circuits.

I. INTRODUCTION

Junction field-effect transistors (JFETs) are depletion-mode semiconductor devices in which channel conduction is controlled by the depletion region formed by a reverse-biased p–n junction. Owing to their simple structure, low gate leakage, and stable analog behavior, JFETs continue to serve as benchmark devices in device physics and low-noise analog circuit design.

As device dimensions scale down, conventional planar JFETs experience degraded electrostatic control, resulting in increased leakage current and short-channel effects. Multi-gate architectures, particularly gate-all-around (GAA) structures, offer superior gate control by completely surrounding the channel, enabling uniform depletion and improved modulation of channel charge. These characteristics make GAA-based devices attractive for nanoscale and low-power analog applications.

This work presents a systematic TCAD-based comparison of planar and GAA n-channel JFETs using consistent geometry definitions, doping profiles, meshing strategies, and physical models.

II. DEVICE STRUCTURE AND SIMULATION METHODOLOGY

A. Planar JFET Structure

The planar n-channel JFET was constructed using Sentaurus Structure Editor (SDE). All dimensions were defined in micrometers, and distinct regions were assigned for the channel,

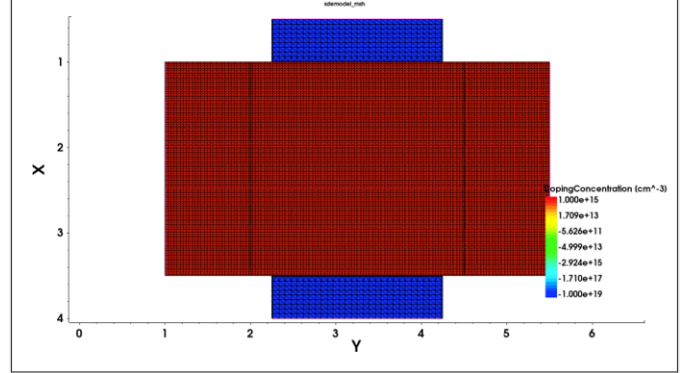


Fig. 1: Cross-sectional schematic of the long-channel planar n-channel JFET (Device 1).

p-type gate, and source/drain extensions. Ohmic contacts were applied to the source, drain, and gate terminals. Uniform constant doping profiles were used across all regions.

B. Mesh Generation

A nonuniform mesh was generated with fine grid spacing in the channel and gate junction regions to accurately resolve depletion effects. The final structure and doping distributions were verified using Sentaurus Visual.

C. Device Simulation

Electrical simulations were performed using Sentaurus SDevice by solving Poisson's equation coupled with electron continuity equations. Doping-dependent mobility, high-field velocity saturation, and bandgap narrowing models were enabled. Identical physical models and solver settings were applied to all devices to ensure a fair comparison.

D. GAA JFET Structure

The GAA JFET consists of a cylindrical silicon channel fully surrounded by a conformal p-type gate region. Source and drain extensions are aligned along the transport direction. Mesh refinement was applied at the channel–gate interface to capture strong electrostatic coupling.

III. RESULTS AND DISCUSSION

The electrical characteristics of the planar and GAA n-channel JFETs were extracted from TCAD simulations.

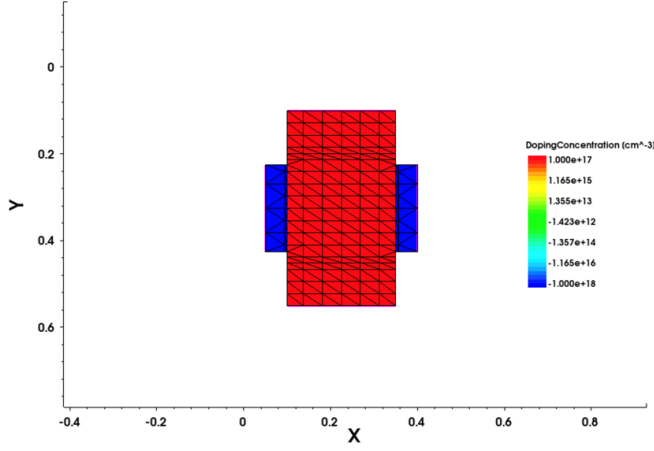


Fig. 2: Cross-sectional schematic of the scaled planar n-channel JFET (Device 2).

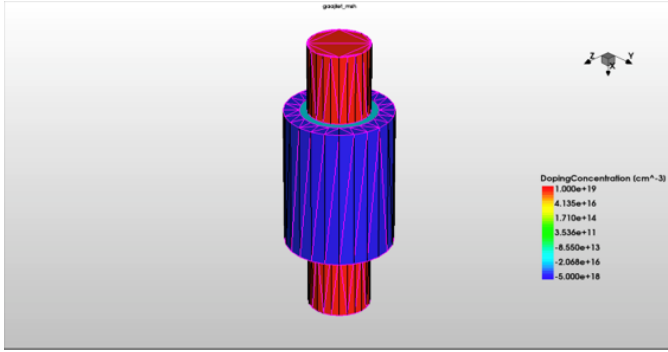


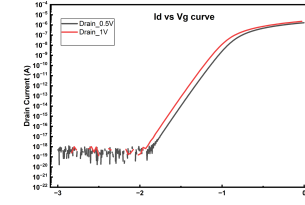
Fig. 3: Three-dimensional schematic of the GAA n-channel JFET.

TABLE I: Comparison of Electrical Performance Metrics

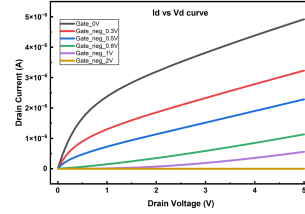
Parameter	Planar	Scaled Planar	GAA
g_m (mS)	1.12	10.36	22.0
I_{ON} (μ A)	1.0	1.0	1.1
I_{OFF} (A)	10^{-19}	10^{-18}	5×10^{-20}
V_{TH} (V)	-0.8	-1.0	-0.65

For planar devices, Device 1 represents a long-channel structure, while Device 2 corresponds to a geometrically scaled version. Device 2 exhibits a peak transconductance of 10.36 mS, nearly an order of magnitude higher than that of Device 1 (1.12 mS) at $V_{DS} = 0.5$ V. Both devices achieve comparable ON-state currents of approximately 1 μ A. However, the OFF-state leakage increases from $\sim 10^{-19}$ A in Device 1 to $\sim 10^{-18}$ A in Device 2, accompanied by a threshold voltage shift from -0.8 V to -1.0 V.

The performance trends summarized in Table I clearly demonstrate the strong influence of gate geometry on electrostatic control in n-channel JFETs. Although planar device scaling enhances transconductance by improving gate-to-channel coupling and reducing effective channel length, it also leads to increased off-state leakage as a result of weakened channel depletion control and enhanced short-channel effects.

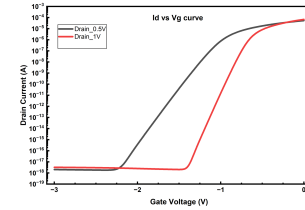


(a) I_D - V_G characteristics

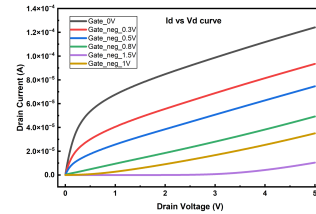


(b) I_D - V_D characteristics

Fig. 4: Electrical characteristics of the long-channel planar JFET (Device 1).



(a) I_D - V_G characteristics



(b) I_D - V_D characteristics

Fig. 5: Electrical characteristics of the scaled planar JFET (Device 2).

In contrast, the GAA JFET benefits from a surround-gate configuration that enables uniform radial depletion of the channel, providing superior electrostatic control over carrier conduction. This improved control results in simultaneously higher transconductance and reduced off-state leakage compared to planar counterparts. Such characteristics are particularly advantageous for low-power analog circuits, where stable biasing, low standby power consumption, and robust operation across process variations are critical. Overall, these results highlight the inherent trade-offs associated with planar JFET miniaturization and demonstrate that GAA architectures offer a more scalable solution for maintaining performance while suppressing leakage in advanced device geometries.

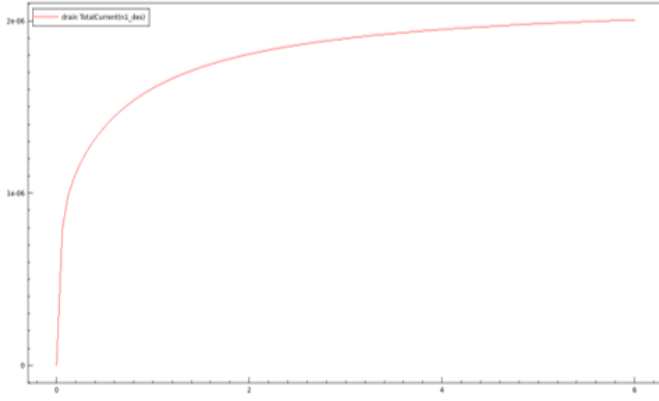


Fig. 6: I_D – V_D characteristics of the GAA JFET at $V_G = -2$ V.

IV. CONCLUSION

In this work, planar and gate-all-around (GAA) JFET structures were designed and simulated using Sentaurus TCAD to study the effects of device geometry and scaling on electrical behavior. The long-channel planar JFET served as a reference device and exhibited stable depletion-mode operation with low leakage and well-defined output and transfer characteristics. Scaling the planar JFET to a shorter channel length resulted in a noticeable improvement in output current modulation and transconductance, as observed from the simulated characteristics, while also introducing increased leakage, highlighting the trade-offs associated with planar device scaling. For the GAA JFET, output characteristics (I_D – V_{DS}) were successfully obtained, demonstrating proper channel control and current saturation behavior under drain bias. Although transfer characteristics were not extracted for the GAA device, the output results confirm the feasibility of the gate-all-around structure and its ability to control channel conduction. Overall, this study establishes a consistent TCAD-based framework for analyzing planar and GAA JFET architectures and provides insight into how device scaling and gate geometry influence JFET performance.

V. CONTRIBUTIONS

- **Sahil** focused on the development and simulation of the initial long-channel planar JFET, serving as the unscaled reference device for comparison, including the extraction and plotting of output and transfer characteristics.
- **Manasi** handled the layout and meshing of the planar JFET structures and performed simulations for the scaled short-channel version, including the extraction and plotting of output and transfer characteristics.
- **Prajwal** was responsible for the design and simulation of the gate-all-around (GAA) JFET structure, including geometry creation, doping definition, and extraction of output characteristics.
- **All authors** jointly contributed to the analysis of simulation results, preparation of figures and tables, and the writing and revision of the final report.

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